

6)UNIVERSITY OF CAPE TOWN
DEPARTMENT OF STATISTICAL SCIENCES - STA2020F: APPLIED STATISTICS
CLASS TEST 2 – 9 May 2025

Time: 90
MINUTES

Max Marks: 50

Internal Ms. A. Cloete
examiners: Mrs. D. van Wyk-
de Ridder

Instructions: Answer ALL questions.
Marks are awarded for intermediate calculations.
Answer all MCQ questions *IN PENCIL* on the MCQ sheet provided.
Answer all written questions *IN PEN* in the answer booklet(s) provided.
Answer each written question on a *NEW PAGE* in your answer book.
ROUND ALL ANSWERS TO 2 DECIMAL PLACES UNLESS OTHERWISE SPECIFIED.
DO NOT ROUND INTERMEDIARY CALCULATIONS.

SECTION A: MCQ QUESTIONS

[12]

1. Consider the following scenario. Two marketing analysts, Alex and Jamie, are debating whether their promotional campaigns differ in performance. Jamie claims that their average sales uplift is the same, while Alex believes they are different.

A third analyst, Rina, decides to join the study afterwards. Each analyst contributes data from 6 promotional campaigns, recording the percentage increase in sales for each. They run an ANOVA to test whether there is a difference in average campaign performance.

The results show an F-statistic of 8.24 and a p-value of 0.007.

Which of the following statements is TRUE? (2)

- A. The results show that at least one analyst's average differs, but we can't say if Alex's differs from Jamie's.
 - B. The result confirms Alex's average is different from both Jamie's and Rina's.
 - C. Jamie is wrong because a significant result means all three averages differ.
 - D. The test proves Alex is incorrect, since the result is not significant.
 - E. None of the above statements is correct.
2. A study was conducted on the taste preferences of caged cockatiels. A sample of birds bred at the University of California, Davis, was randomly divided into three experimental groups using a Completely Randomised Design. Group 1 was fed purified water in bottles on both sides of the cage. Group 2 was fed water on one side and a liquid sucrose (sweet) mixture on the opposite side of the cage. Group 3 was fed water on one side and a liquid sodium chloride (sour) mixture on the opposite side of the cage. The variable of interest to the researchers was the total

consumption of liquid by each cockatiel. What is the experimental unit, treatment factor(s), factor(s) levels and response of this study? (2)

- A. Experimental unit: Each cockatiel, Factor: Type of liquid offered, Factor levels: Water, Water & Sucrose, Water & Sodium Chloride, Response: Total liquid consumed.
 - B. Experimental unit: A cage, Factor: Type of liquid offered and Liquid position, Factor levels: Water, Sucrose, Sodium Chloride (Type of Liquid), Left vs Right (Side), Response: Total liquid consumed.
 - C. Experimental unit: A group of cockatiels, Factor: Type of liquid offered, Factor levels: Water vs Water, Water vs Sucrose, Water vs Sodium Chloride, Response: Total liquid consumed.
 - D. Experimental unit: Each cockatiel, Factor: Liquid position, Factor levels: Left vs Right, Response: Total liquid consumed.
 - E. Experimental unit: Each cockatiel, Factor: Type of liquid offered, Factor levels: Water vs Water, Water vs Sucrose, Water vs Sodium Chloride, Response: Preferred Liquid.
3. Which of the following statements best explains why Spearman correlation is less sensitive to outliers than Pearson? (2)
- A. It uses squared deviations, which causes large values to have a greater influence on the result, reducing the impact of small changes.
 - B. It does not assume equal variance, which stabilises the influence of extreme values across the range of data.
 - C. It uses ranks instead of raw values, so extreme data points are treated according to their position in order, not their magnitude.
 - D. It does not require normality, ensuring that the effect of extreme values is minimised.
 - E. It ignores high values in the data, focusing only on mid-range ranks.
4. Classical decomposition is still widely used for forecasting, but there are some drawbacks to using this method. Choose the correct answer from the drawbacks listed below. (2)
- A. In a small number of time periods, you might come across particularly unusual values that classical decomposition can only model through simple linear regression, which tends to overcomplicate matters.
 - B. Classical decomposition requires a great extent of variation in seasonality to estimate the seasonal component accurately.
 - C. Due to moving average smoothing, the trend estimate does not include the first and last few observations of the time series.
 - D. The trend estimate in classical decomposition tends to over-complicate rises and falls in the data.
 - E. Moving average smoothing is challenging to apply in the classical decomposition framework. You can only ever use a centred moving average process.

Refer to the TIME SERIES APPENDIX: PART 1: DATA to answer the following question.

5. An MA(B) process was fitted to the time series on the average price of a dozen eggs in South Africa observed over a specific period in 2024. Choose the correct answer. (2)

- A. **B** = 3 and, rounded to the second decimal, the value of **C** = 21.49.
- B. **B** = 7 and, rounded to the second decimal, the value of **D** = 20.14.
- C. **B** = 4 and, rounded to the second decimal, the value of **D** = 20.11.
- D. **B** = 12 and, rounded to the second decimal, the value of **D** = 19.89.
- E. **B** = 7 and, rounded to the second decimal, the value of **E** = 22.51.

Refer to the **TIME SERIES APPENDIX: PART 2: TIME PLOT** to answer the following question.

6. Suppose you wanted to forecast the best short-term future values for the time series on home loan interest rates (called *interest_rate*) in RStudio. The correct code would be: (2)

- A. `naive(interest_rate, h=102)`
- B. `snaive(interest_rate, h=8)`
- C. `snaive(interest_rate, h=36)`
- D. `rwf(interest_rate, h=8, drift = TRUE)`
- E. `rwf(interest_rate, h=48, drift = TRUE)`

SECTION B: ANOVA

[16]

QUESTION 1

[6]

To investigate whether the type of telescope used affects the number of stars an observer can count in the same fixed region of the night sky, a set of 15 independent observers is each randomly assigned to one of the three telescope types:

- Type A: Small amateur telescope
- Type B: Mid-sized research telescope
- Type C: Large professional observatory telescope

All observers count the number of stars they can see in a fixed patch of sky at the same time of night under similar weather conditions. The data is shown below, together with other statistics that have been calculated.

Type A	Type B	Type C
28	45	62
30	42	65
27	47	63
29	43	67
31	46	64

Total sums of squares = 3152.93, Residual sums of squares = 42, Overall mean = 45.93

- a. Write down the appropriate model equation and state what each term represents. (2)

$$Y_{ij} = \mu + A_i + e_{ij}$$

Where Y_{ij} is the count of stars for observer j using telescope type i

μ is the overall mean count

A_i is the effect of the i th telescope type

e_{ij} are the error terms

[0.5] for the correct model equation

[0.5] for saying what each term represents in general words

[1] for stating what each term represents in context of the experiment

- b. What are the fitted values of this model? Give the actual numeric values for Type A and B. (1)

The fitted values are the treatment means:

Type A = 29 (✓ 0.5)
Type B = 44.6 (✓ 0.5)

- c. What is the standard error of a treatment mean in this experiment? (1)

$$MSE = RSS/(N - a) = 42/(15-3) = 3.5 \quad (\checkmark 0.5)$$

$$SE = \sqrt{MSE/n} = \sqrt{3.5/5} = 0.836660026534 \approx 0.84 \quad (\checkmark 0.5)$$

- d. Calculate a 95% confidence interval for the mean count of stars when using Type A telescopes. (1)

$$\bar{x} = 29$$

$$t_{12, \alpha/2} = 2.179 \quad (\checkmark 0.5)$$

$$se = 0.84$$

$$\bar{x} \pm 2.179 \times 0.84 = (27.17, 30.83) \quad (\checkmark 0.5)$$

Mark with error from the previous two questions. If they don't round the 0.84 SE, the interval is (27.18, 30.82). If they use the wrong t value, 0.5 only.

- e. Calculate the F-statistic for testing the null hypothesis that telescope type has no effect. (1)

$$SS_{\text{Treat}} = SST - SSR = 3152.93 - 42 = 3110.93 \quad (\checkmark 0.5)$$

$$MS_{\text{Treat}} = 3110.93/2 = 1555.465$$

$$MSE = 42/12 = 3.5$$

$$(\checkmark 0.5) \text{ for both MS}$$

$$F = 1555.465 / 3.5 = 444.4186 = 444.42 \quad (\checkmark 1)$$

QUESTION 2

[6]

A researcher is studying the effects of three types of fiscal policy interventions on consumer spending. The interventions are: Direct cash transfers, Tax cuts, and Government spending programs. Each intervention is implemented in four distinct geographic areas (North, South, West, and East). After a month, average monthly consumer spending (in R1000s per person) is recorded for each case. The study aims to determine whether fiscal policy type has a significant effect on consumer spending.

Complete the following incomplete ANOVA table and identify the type of design used.

Source	Df	Sums of Squares	Mean Square	F
Region		0.47		
Policy			0.27	
Residuals		0.21		
Total	11			

- a. Identify the design and type of factors. (2)

This is a Randomised Complete Block Design / RCBD (☑ 1). Region is the blocking factor (☑ 0.5) and Policy is the treatment factor (☑ 0.5).

- b. Copy and complete the ANOVA table. **Round all values to two decimal places before using them in subsequent calculations.** (4)

Source	Df	Sums of Squares	Mean Square	F
Region	3	0.47	0.16	4
Policy	2	0.54	0.27	6.75
Residuals	6	0.21	0.04	
Total	11	1.22		

(☑ 1) mark for DF column

(☑ 1) mark for sums of squares

(☑ 1) mark for MSE

(☑ 1) mark for F values

QUESTION 3

[4]

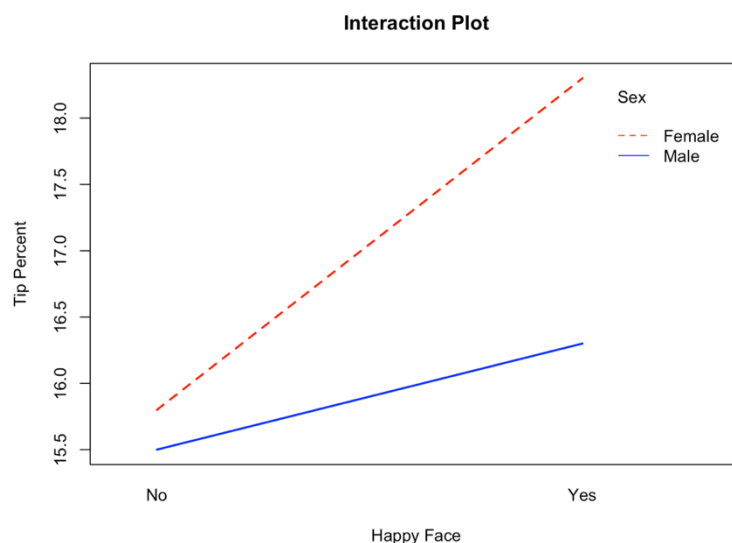
Is there a financial incentive when a restaurant server writes a friendly note or draws a “happy face” on your restaurant check, or is this just a friendly act? Temple University psychologists conducted a randomised experiment to investigate whether drawing a happy face on the back of a restaurant bill increased the average tip to the server (Rind and Rhinda, 1996). They were interested in the effect of drawing a happy face on the back of the bill and whether this effect depends on the server’s biological sex (using a 10% significance level). The following R output was generated.

> summary(model)

```

              Df Sum Sq Mean Sq  F value    Pr(>F)
HappyFace      1  27.225   27.225  16.25373 0.0002749
Sex            1  13.225   13.225   7.89552 0.0079631
HappyFace:Sex  1   7.225    7.225   4.31343 0.0450086
Residuals     36  60.300    1.675
---

```



Decide whether there is a significant interaction (provide a reason) and if so, identify and interpret the difference(s) in mean response if the relevant effects are present. (4)

The p-value is smaller than the significance level (☒ 1), which means we have evidence against the null hypothesis of no interaction (☒ 1).

Interaction: The increase in mean tip percentage is more pronounced for female servers than for male servers (☒ 1☒ 1).

1 mark saying there is an interaction

1 mark for reason (p-value smaller)

2 marks for correct interpretation of interaction

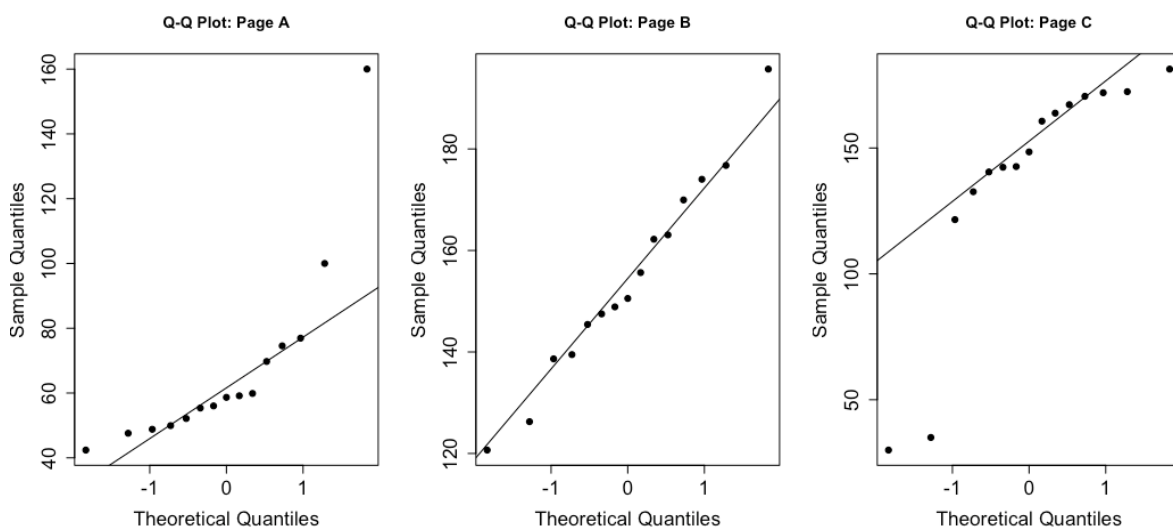
SECTION C: NON-PARAMETRIC TESTS

[8]

QUESTION 1

[2]

A data analyst is investigating whether the average time spent on a company's website differs across three different landing page designs: Page A, Page B, and Page C. For each version, 15 users were randomly assigned a page design, and the time they spent on the website (in seconds) was recorded. Based on the plots she produces, she reckons that a non-parametric test is the most appropriate.



Comment on the analysis performed. Was using a non-parametric test appropriate? Provide a reason(s) for your answer and identify the appropriate test she should use. (2)

Yes, a non-parametric test is appropriate here, since the Q-Q plots show deviation from normality (there is extreme deviation at the tails in at least two of the plots) (☒ 1).

Or students could say the Q-Q plots show outliers in the first and third groups.

The appropriate test would be the Kruskal-Wallis test (☒ 1).

QUESTION 2

[6]

In a study on sweet potato yields, data were collected over 7 agricultural seasons across three different zones. The researchers are interested in differences between zones as they are believed to differ in their suitability for growing sweet potatoes, with Zone I being the least favourable and Zone III being the most favourable.

The yields (in tons per hectare) for each zone over the seven seasons are given in the table below:

Zone	1	2	3	4	5	6	7
Zone I	4.17	2.30	7.93	4.18	4.18	6.16	5.69
Zone II	5.91	6.03	7.69	5.78	5.78	8.45	6.68
Zone III	6.35	7.52	6.30	8.79	8.78	8.08	11.00

a. What is the appropriate non-parametric test for this data? (1)

Friedman test (✓ 1)

b. Perform the appropriate test using a 10% significance level. Marks will be awarded for:

- Stating the correct null and alternative hypotheses. (1)
- Calculating the value of the test statistic. (2)
- Making the correct decision and conclusion. (2)

Null: There is no difference in sweet potato yield among the three zones (✓ 0.5).

Alt: There is a difference in sweet potato yield among the three zones (✓ 0.5).

Zone	1	2	3	4	5	6	7	
Zone I	4.17 1	2.30 1	7.93 3	4.18 1	4.18 1	6.16 1	5.69 1	9
Zone II	5.91 2	6.03 2	7.69 2	5.78 2	5.78 2	8.45 3	6.68 2	15
Zone III	6.35 3	7.52 3	6.30 1	8.79 3	8.78 3	8.08 2	11.00 3	18

(✓ 1) mark correct ranks (check rank sums)

$n = 7$

$k = 3$

$$\begin{aligned}
 F_r &= \frac{12}{n k (k + 1)} \sum_{j=1}^{\{k\}} R_j^2 - 3n(k + 1) \\
 &= \frac{12}{7 \times 3 \times (4)} \sum_{j=1}^{\{k\}} (9^2 + 15^2 + 18^2) - 3 \times 7 \times (4) \\
 &= 6
 \end{aligned}$$

(✓ 1) mark for correct test stat, full (✓ 2) marks for just stating the correct test stat.

Critical value: 4.605

P-value is approximately 0.05

The test statistic is larger than the critical value, **or** the p-value is smaller than the significance level (✓ 0.5). Therefore, we reject the null hypothesis (✓ 0.5) and conclude that the evidence suggests a difference in yield between the zones (✓ 1).

Wrong test conducted, maximum marks received is 3: Hypotheses are correct for the Friedman (✓ 1) and then if conclusion is right based on wrong test stat (✓ 1✓ 1)

SECTION D: TIME SERIES ANALYSIS

[14]

QUESTION 1

[10]

The marketing director at Eggcellent Eggs tasked a team of data scientists to track and investigate the daily price of eggs over approximately four months (122 days) in 2024. She wanted to understand where the market was headed for budgetary purposes. The team prepared some graphs and notes in **TIME SERIES APPENDIX: PART 1** for her to review.

- a. Give a detailed description of the time series components (observed and unobserved) from the time plot of the average price for a dozen eggs in South Africa recorded over the mentioned time period in 2024. (3)

Trend: The trend is increasing (☒ 0.5) and linear (☒ 0.5) **OR** non-linear (☒ 0.5) since the unusual observations in May prove to influence the trend quite significantly, as we see from the simple linear regression output for the trend estimates.

Seasonality: There is seasonality (☒ 0.5), and the length of the seasonal variation is one week or 7 days (☒ 0.5). No marks for 'daily' observations; it's given in the heading of the plot.

Cyclical: No cyclical present - the observations in May can be described as an unusual event with some outlying values (☒ 0.5) **OR** Cyclical present – the three weeks in May have a clear deviation from the overall trend and it lasts for more than two weeks (☒ 0.5). *Four experts consulted – two voted for presence of outliers and two voted for cyclical!*

Random variation: Random variation is present (☒ 0.5).

- b. i) Determine the forecasted value for time $t = 124$ with seasonally-adjusted trend estimates in the forecast model determined from classical decomposition. (2)

$$\hat{Y}_{124} = \hat{T}_{124} \times \hat{S}_{124} = (\hat{\beta}_0 + \hat{\beta}_1(124)) \times \hat{S}_5 = (19.72621 + 0.02932(124)) \times 0.9257893 = 21.62879 \approx 21.63$$

- (p0.5) for the correct intercept (still award the mark if they rounded to 19.73).
- (p0.5) for the correct slope (still award the mark if they rounded to 0.03).
- (p0.5) for the correct SI used (still award the mark if they rounded to 0.93).
- (p0.5) for the correct final answer (still award the mark if rounding leads to 21.81).
- Lose 0.5 marks on the total if students calculate the unadjusted SI from the data to use in the expression for the classical decomposition model. Students need to use the adjusted SI provided.

- ii) Do you think that this specific classical decomposition model would provide good estimates of future values? Refer to the R output related to the trend estimate and time plot to provide a reason(s) for your answer. (2)

No, this classical decomposition model would not provide the best estimates of future values. The R^2 (or adjusted R^2) values indicate that a linear model is not a good fit for the trend estimate (☒ 1) since only about 53% of the variation in the deseasonalized

data is explained by the regression model. When looking at the time plot, there are unusual observations in the weeks of May (☑ 1) that the classical decomposition model is not robust enough to accurately account for.

- Students do not have to interpret the R^2 value, only acknowledge that the fit for the trend estimate is not good.
- Students need to refer to what they observe in the time plot regarding the observations in May to explain why they would expect the classical decomposition model to fail for this time series data. NO vague, general reasons against classical decomposition (see slide 101) are to be accepted as correct.
- If students mention autocorrelation, they can get (☑ 0.5).
- Mention of p-values here are incorrect; the model's predictors are found to be statistically related to the outcome, but the model still doesn't explain much of the outcome's variance. No marks.

- c. Calculate the value of $\mathbf{A} = \hat{S}_4$ (as part of the *decomp_model\$figure* output) and interpret this adjusted seasonal index with as much detail as possible. (2)

$$A = 7 - 1.0613644 - 1.1041016 - 1.0272209 - 0.9257892 - 0.9490082 - 0.9821161 = \mathbf{0.9503996} \approx \mathbf{0.95} \text{ (☑ 0.5)}$$

$\hat{S}_4 = 0.9503996$ (approximated to 0.95) implies that, on average, egg prices on the **fourth day of the week (or Sundays, or 'season 4')** (☑ 0.5) are $100 \times |1 - 0.9503996| = \mathbf{4.96\%}$ (or **5%**) (☑ 0.5) **below** (☑ 0.5) the weekly average egg prices per dozen.

- If students don't include 'average', subtract 0.5 marks.
- If students say 'yearly' or 'monthly' instead of 'weekly', subtract 0.5 marks.

- d. Which simple forecasting method will provide the most accurate long-term forecasts? Identify the corresponding letter (**F**, **G**, or **H**) and name the associated method. Explain why you expect this simple forecasting method to be the most suitable. (1)

H = drift method (☑ 0.5) : The drift method considers the increasing linear trend (☑ 0.5) when forecasting.

- Some students identified F, G and H. Award the mark if they get H correct.
- If students identify H incorrectly, but get the reasoning right, award half a mark.

QUESTION 2

[4]

The Money4You banking group is reviewing its home loan interest rates between 1970 and 2007. Their analysts have found some features that they would like to investigate further, but a preliminary set of observations is presented to the CFO in **TIME SERIES APPENDIX: PART 2**.

- a. Describe in as much detail as possible how you would transform the non-stationary time series on home loan interest rates towards achieving stationarity. Identify the aspects causing the time series to be non-stationary and explain how its effects can be neutralised. (3)

The time series exhibits diverging variation (not constant) (✓ 0.5). The variance can be stabilised by applying a natural logarithm or a power transformation to the time series (✓ 0.5 for EITHER mentioning the natural logarithm OR the power transformation OR Box-Cox transformations).

The time series exhibits a trend, which indicates that the mean is not constant (✓ 0.5). We can detrend the time series via first-order differencing (✓ 0.5). Since there is an aspect of seasonality (✓ 0.5), we will have to difference at lag $g = 4$ (✓ 0.5), to account for the quarterly data under consideration.

- No marks for only mentioning 'detrending' or 'deseasonalizing'.
- First (✓ 0.5) can also be awarded if students just start with 'to stabilise the variation...'.
No marks for a vague 'lag k' reference for differencing.
- No marks for stating that a stationary time series requires a constant mean and variance – I want to know whether the student sees these aspects in the time series under consideration.

- b. Investigate the autocorrelation plot of the transformed data. What do you observe, and what is the implication of your observation(s)? (1)

There is still significant autocorrelation, visible at lags 4 and 18, even after transforming the data (✓ 0.5). Removing the trend and stabilising the variance did not account for all the information in the time series (✓ 0.5), and we need to investigate the time series some more.

- No marks for 'outliers' outside of the 95% CI.
- No marks for saying we 'over-differenced'.
- No marks for saying 'there is autocorrelation between the values' - too vague.
- 0.5 marks for saying there are autocorrelations outside of the CI.
- No marks for saying there is an 'extreme value' at lag 4.
- Strict with wording: students seem to have a general idea, but not the vocabulary.

THE END

TIME SERIES APPENDIX

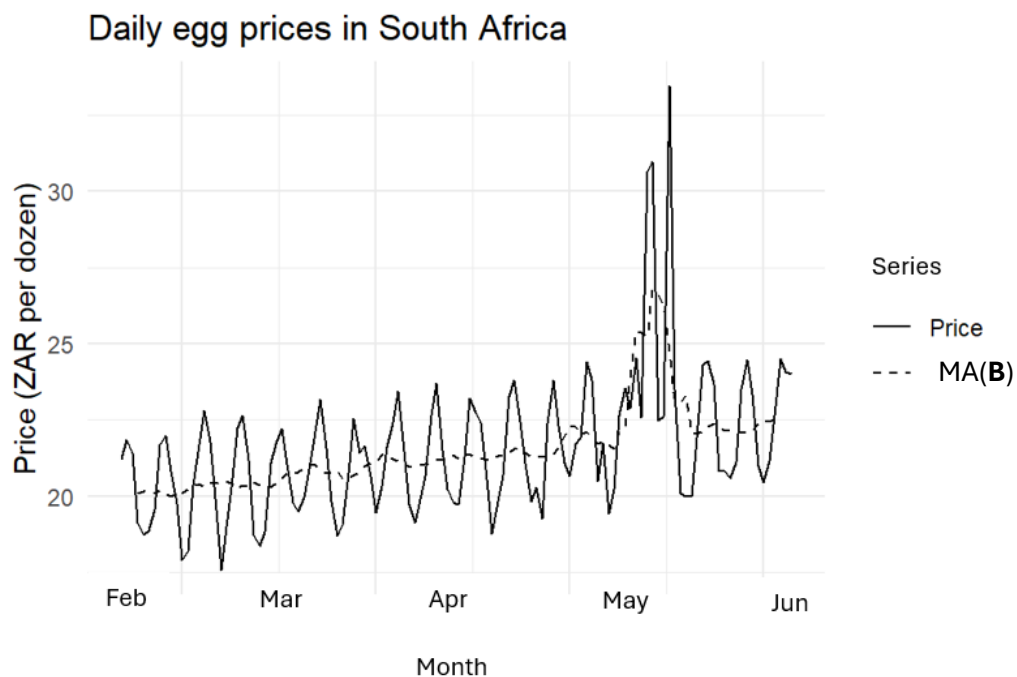
PART 1:

DATA:

Note: T = 122 and 2024/02/01 was a Thursday

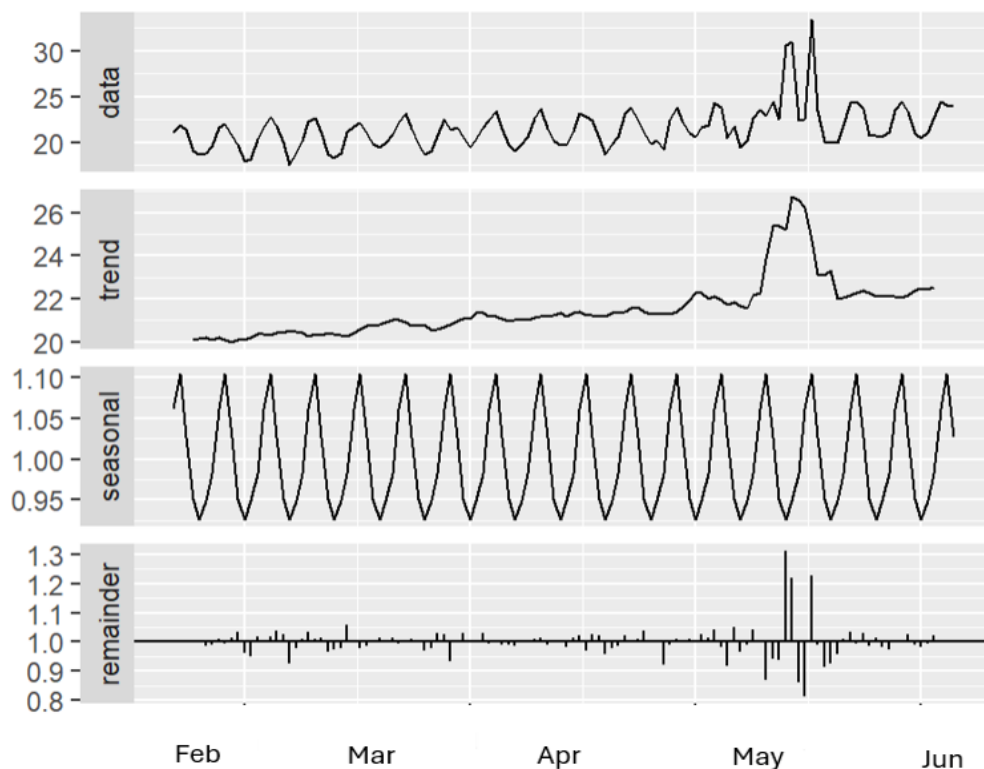
	Date	Price	MA(B)
1	2024/02/01	21.21	
2	2024/02/02	21.86	C
3	2024/02/03	21.41	
4	2024/02/04	19.12	
5	2024/02/05	18.74	
6	2024/02/06	18.86	
7	2024/02/07	19.55	
8	2024/02/08	21.69	
9	2024/02/09	21.99	D
10	2024/02/10	20.82	
11	2024/02/11	19.75	
12	2024/02/12	17.89	
13	2024/02/13	18.2	
14	2024/02/14	20.35	
15	2024/02/15	21.67	
...	...	...	...
112	2024/05/22	21.12	
113	2024/05/23	23.52	
114	2024/05/24	24.46	
115	2024/05/25	23.34	
116	2024/05/26	21	
117	2024/05/27	20.47	E
118	2024/05/28	21.15	
119	2024/05/29	22.64	
120	2024/05/30	24.51	
121	2024/05/31	24.07	
122	2024/06/01	24.01	

TIME PLOT:



CLASSICAL DECOMPOSITION:

Decomposition of multiplicative time series



```
> decomp_model$figure
```

```
[1] 1.0613644 1.1041016 1.0272209 A 0.9257893 0.9490082 0.9821161
```

Call:

```
lm(formula = trend ~ time)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.9962	-0.4524	-0.1539	0.1301	4.1809

Coefficients:

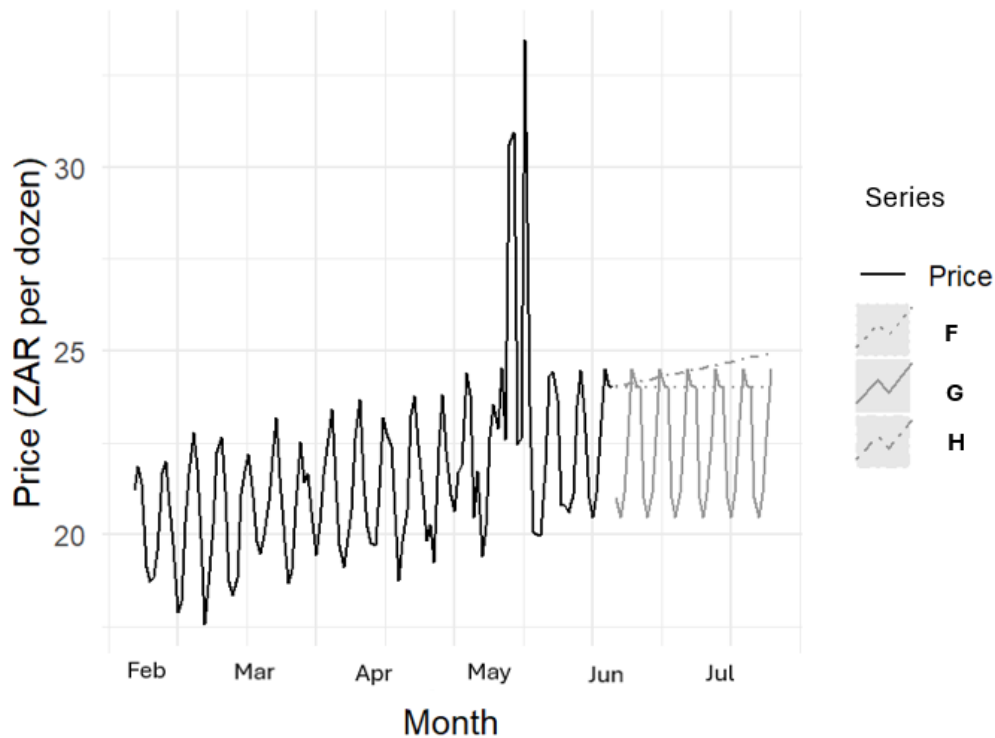
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	19.72621	0.17788	110.90	<2e-16 ***
time	0.02932	0.00254	11.54	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9161 on 114 degrees of freedom
Multiple R-squared: 0.5389, Adjusted R-squared: 0.5349
F-statistic: 133.2 on 1 and 114 DF, p-value: < 2.2e-16

SIMPLE FORECASTING:

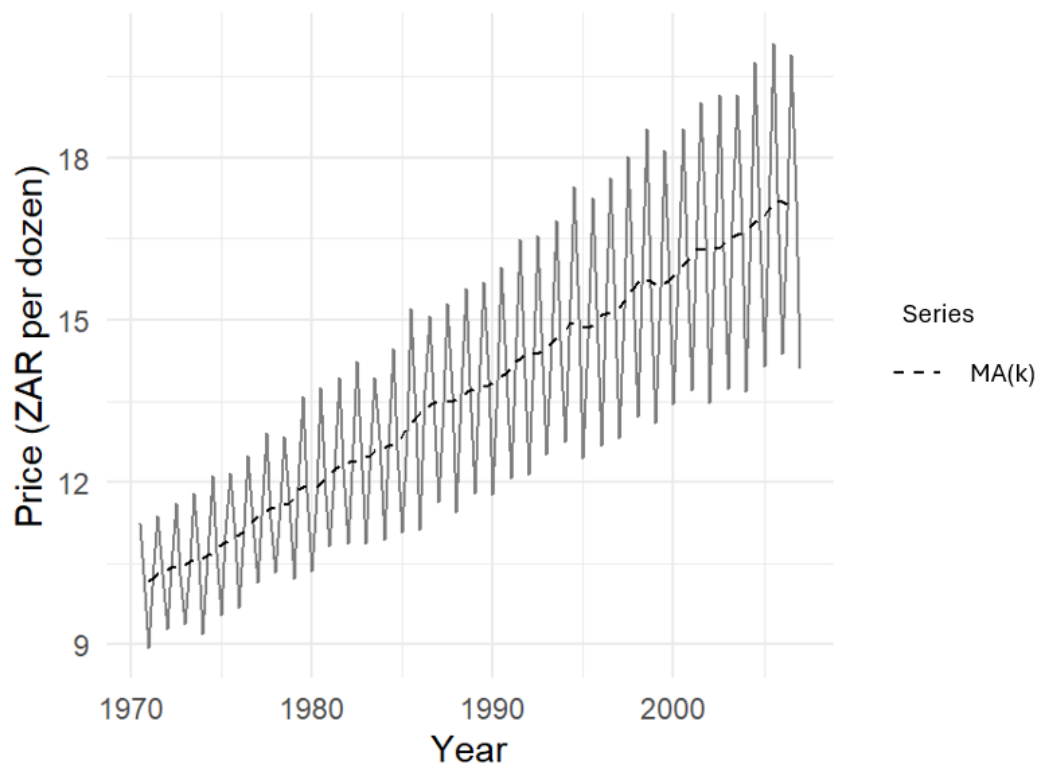
Daily egg prices in South Africa



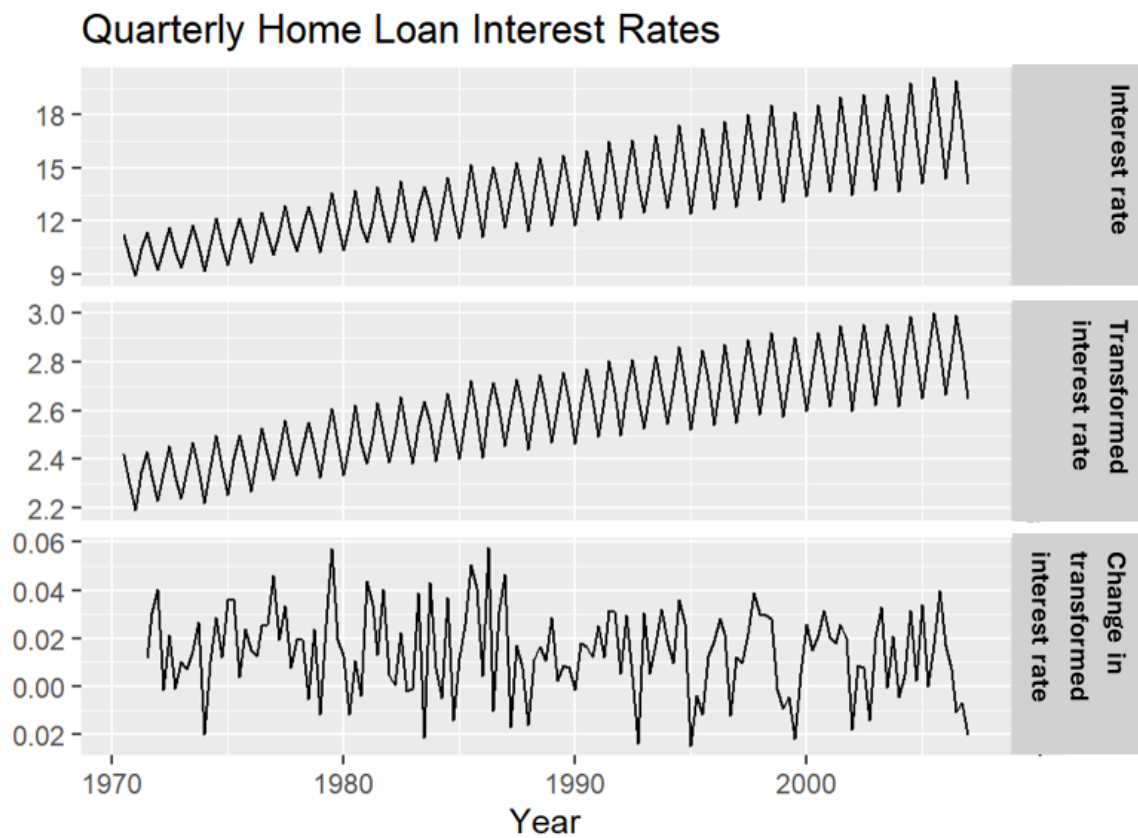
PART 2:

TIME PLOT:

Quarterly Home Loan Interest Rates

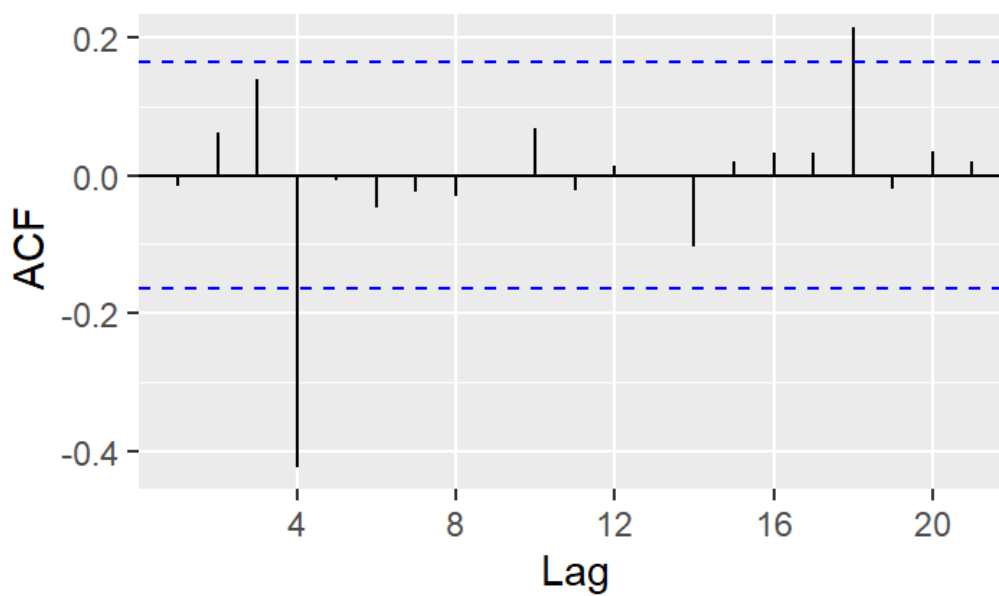


TRANSFORMATION:



AUTOCORRELATION:

Autocorrelation plot of transformed interest rates:



STA2020F: Important Formulae

ANOVA

One-way ANOVA

$$SS_{Total} = \sum_i \sum_j (Y_{ij} - \bar{Y}_{..})^2$$

$$Total\ df = N - 1$$

$$SS_{treatment} = \sum_i r(\bar{Y}_{i.} - \bar{Y}_{..})^2$$

$$Treatment\ df = a - 1$$

$$SSE = \sum_i \sum_j (Y_{ij} - \bar{Y}_{i.})^2$$

$$Error\ df = N - a$$

Two-way ANOVA without interaction

$$SS_{Total} = \sum_i \sum_j (Y_{ij} - \bar{Y}_{..})^2$$

$$Total\ df = N - 1$$

$$SS_{treatment} = \sum_i b(\bar{Y}_{i.} - \bar{Y}_{..})^2$$

$$Treatment\ df = a - 1$$

$$SS_{block} = \sum_j a(\bar{Y}_{.j} - \bar{Y}_{..})^2$$

$$Blocking\ df = b - 1$$

$$SSE = \sum_i \sum_j (Y_{ij} - \bar{Y}_{i.} - \bar{Y}_{.j} + \bar{Y}_{..})^2$$

$$Error\ df = (a - 1)(b - 1)$$

One-way ANOVA with interaction

$$SS_{Total} = \sum_i \sum_j \sum_k (Y_{ijk} - \bar{Y}_{...})^2$$

$$Total\ df = N - 1$$

$$SS_A = \sum_i rb(\bar{Y}_{i..} - \bar{Y}_{...})^2$$

$$A\ df = a - 1$$

$$SS_B = \sum_j ra(\bar{Y}_{.j.} - \bar{Y}_{...})^2$$

$$B\ df = b - 1$$

$$SS_{AB} = \sum_i \sum_j r(\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...})^2$$

$$AB\ df = (a - 1)(b - 1)$$

$$SSE = \sum_i \sum_j \sum_k (Y_{ijk} - \bar{Y}_{ij.})^2$$

$$Error\ df = abr - 1$$

NON-PARAMETRIC

Friedman Statistic:

$$F_r = \left[\frac{12}{bk(k+1)} \sum_{j=1}^k T_j^2 \right] - 3b(k+1), \text{ df} = k - 1$$

Kruskal Wallis Statistic:

$$H = \left[\frac{12}{n_T(n_T+1)} \sum_{i=1}^k \frac{T_i^2}{n_i} \right] - 3(n_T + 1), \text{ df} = k - 1$$

Mann-Whitney-Wilcoxon Test Statistic (normal approximation, n_1 and/OR $n_2 \geq 10$):

$$z = \frac{T - \mu_T}{\sigma_T} \text{ where } \mu_T = \frac{n_1(n_1+n_2+1)}{2}, \sigma_T = \sqrt{\frac{n_1 n_2 (n_1+n_2+1)}{12}}$$

Spearman Rank correlation ($n < 10$):

$$r_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

Spearman Rank correlation (normal approximation, $n \geq 10$):

$$z = \frac{r_s - 0}{\sqrt{\frac{1}{n-1}}} = r_s \sqrt{n-1}$$

Wilcoxon Signed Rank Sum Statistic (normal approximation, $n \geq 10$):

$$z = \frac{T - \mu_T}{\sigma_T} \text{ where } \mu_T = 0, \sigma_T = \sqrt{\frac{n(n+1)(2n+1)}{6}}$$

TIME SERIES FORMULAE

CLASSICAL DECOMPOSITION MODEL

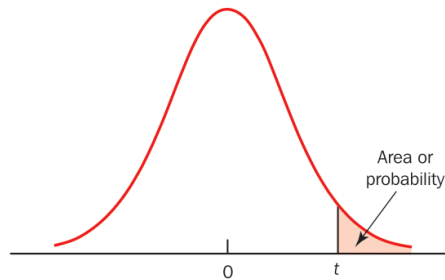
Multiplicative model formulation: $Y_t = T_t \times S_t \times R_t$ assuming $C_t = 1$ i.e. it is negligible

Trend Equation:

$$T_t = \hat{\beta}_0 + \hat{\beta}_1 t$$

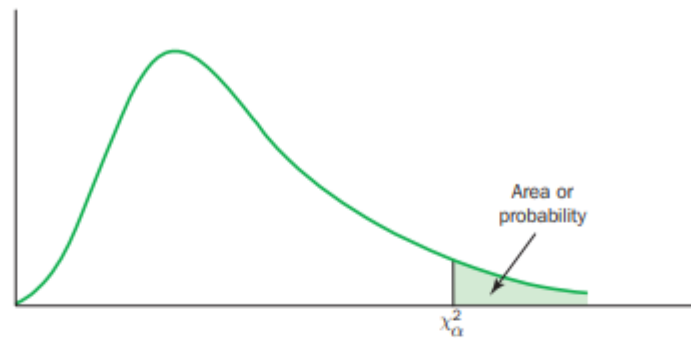
Forecast equation:

$$\hat{Y}_t = (\hat{\beta}_0 + \hat{\beta}_1 t) \times SI_t$$

TABLE 2 *t* distribution

Entries in the table give *t* values for an area or probability in the upper tail of the *t* distribution. For example, with ten degrees of freedom and a .05 area in the upper tail, $t_{.05} = 1.812$.

Degrees of freedom	Area in upper tail					
	.20	.10	.05	.025	.01	.005
1	1.376	3.078	6.314	12.706	31.821	63.656
2	1.061	1.886	2.920	4.303	6.965	9.925
3	.978	1.638	2.353	3.182	4.541	5.841
4	.941	1.533	2.132	2.776	3.747	4.604
5	.920	1.476	2.015	2.571	3.365	4.032
6	.906	1.440	1.943	2.447	3.143	3.707
7	.896	1.415	1.895	2.365	2.998	3.499
8	.889	1.397	1.860	2.306	2.896	3.355
9	.883	1.383	1.833	2.262	2.821	3.250
10	.879	1.372	1.812	2.228	2.764	3.169
11	.876	1.363	1.796	2.201	2.718	3.106
12	.873	1.356	1.782	2.179	2.681	3.055
13	.870	1.350	1.771	2.160	2.650	3.012
14	.868	1.345	1.761	2.145	2.624	2.977
15	.866	1.341	1.753	2.131	2.602	2.947
16	.865	1.337	1.746	2.120	2.583	2.921
17	.863	1.333	1.740	2.110	2.567	2.898
18	.862	1.330	1.734	2.101	2.552	2.878
19	.861	1.328	1.729	2.093	2.539	2.861
20	.860	1.325	1.725	2.086	2.528	2.845
21	.859	1.323	1.721	2.080	2.518	2.831
22	.858	1.321	1.717	2.074	2.508	2.819
23	.858	1.319	1.714	2.069	2.500	2.807
24	.857	1.318	1.711	2.064	2.492	2.797
25	.856	1.316	1.708	2.060	2.485	2.787
26	.856	1.315	1.706	2.056	2.479	2.779
27	.855	1.314	1.703	2.052	2.473	2.771
28	.855	1.313	1.701	2.048	2.467	2.763
29	.854	1.311	1.699	2.045	2.462	2.756

TABLE 3 Chi-squared distribution

Entries in the table give χ^2_{α} values, where α is the area or probability in the upper tail of the chi-squared distribution. For example, with ten degrees of freedom and a .01 area in the upper tail, $\chi^2_{0.1} = 23.209$

Degrees of freedom	Area in upper tail									
	.995	.99	.975	.95	.90	.10	.05	.025	.01	.005
1	.000	.000	.001	.004	.016	2.706	3.841	5.024	6.635	7.879
2	.010	.020	.051	.103	.211	4.605	5.991	7.378	9.210	10.597
3	.072	.115	.216	.352	.584	6.251	7.815	9.348	11.345	12.838
4	.207	.297	.484	.711	1.064	7.779	9.488	11.143	13.277	14.860
5	.412	.554	.831	1.145	1.610	9.236	11.070	12.832	15.086	16.750
6	.676	.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.647	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.041	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.558
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.878	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645

(continued)